

Immune-cell RNA-seq: donor adjustment and target ranking

QV.23.100 | Corvenlith Therapeutics | 18 Dec 2023

v3 FINAL_final | Dr T. Zawiernicki

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Biological question and decision

- Which pathway remains reproducible after donor adjustment?
- Donor-balanced analysis reduced the apparent inflammatory signature.
- Prioritise the reproducible receptor pathway for follow-up.
- This is Omics target prioritisation, not proof of receptor causality.

Engagement provenance

- Immune-cell RNA-seq supplied for QV.23.100.
- Accession QDATA2023-0617 anchors input provenance.
- Proposal dated 12 Sep 2023; analysis started 30 Oct 2023.
- Interim readout 23 Nov 2023; final model gate 18 Dec 2023.

Revision note: v3 FINAL_final

- This revision puts donor-adjusted pathway prioritisation before the inflammatory interpretation.
- The inflammatory signature is retained as a qualified observation, not the progression recommendation.
- The receptor-pathway recommendation is the final conclusion of this saved readout.

Data acceptance

- Reconcile donor labels, sample identifiers and expression inputs.
- Check library-level distributions before fitting biological contrasts.
- Preserve missing metadata and exclusions in the analysis notebook.

Donor structure matters

- Donor-associated expression can resemble a treatment or disease-associated signature.
- Repeated material from a donor is not independent biological replication.
- Keep donor identity attached to expression inputs through modelling.

Unadjusted interpretation

- The initial interpretation emphasised an apparent inflammatory signature.
- Without donor adjustment, donor composition can influence pathway ordering.
- Retain this view as the comparator, rather than the final target ranking.

Adjusted analysis question

- Estimate the biological contrast while accounting for donor structure.
- Keep the accepted input set and contrast definition explicit.
- Do not confound a changed model specification with newly generated experimental evidence.

Donor-balanced comparison

- Use donor-balanced evidence to reassess the pathway interpretation.
- Compare adjusted and unadjusted direction of effect.
- Flag conclusions whose support depends on donor representation.

What changed

- The apparent inflammatory signature weakened after donor-balanced analysis.
- The change reduces its priority as a general biological explanation.
- Weakening does not demonstrate absence of inflammatory biology.

What remains reproducible

- The receptor pathway remains the reproducible priority.
- Carry the pathway into a focused biological follow-up.
- No specific receptor identifier is assigned without the source target card.

Pathway comparison at the gate

- Inflammatory interpretation: retain with donor-dependence caveat.
- Receptor pathway: progress on the reproducible adjusted interpretation.
- Other candidate explanations: do not imply a rank unsupported by the supplied outcome.

Reporting conventions

- Report expression contrasts as log2 fold change with adjusted probabilities.
- Keep pathway definitions and annotation versions alongside contrasts.
- This readout contains the qualitative decision; numerical pathway tables are not supplied.

Target-card requirements

- Link the receptor-pathway target card to the adjusted contrast.
- Record biological rationale separately from statistical association.
- Carry donor sensitivity and assay feasibility into the follow-up decision.

Alternative explanations

- Cell composition may contribute to pathway-level expression.
- Correlated donor and biological terms limit attribution.
- A reproducible pathway signal need not identify the direct intervention point.

Follow-up question

- Does a targeted experiment support receptor-pathway involvement?
- Define the perturbation and functional readout before observing the outcome.
- Use donor-aware sampling in the follow-up design.

Follow-up design boundaries

- Balance donor contribution across experimental conditions.
- Include controls that distinguish functional response from viability loss.
- Retain expression evidence as context for the functional readout.

Model uncertainty to retain

- Residual donor effects may remain.
- Pathway membership is annotation dependent.
- Association does not establish therapeutic benefit or direction of intervention.

Reproducibility responsibilities

- Dr E. Ostlund: donor-aware model interpretation.
- Nadia Rahmani: annotation and donor-to-sample mapping.
- Dr T. Zawiernicki: target-prioritisation recommendation.

Archive handover

- Keep input mapping and model definitions under QDATA2023-0617.
- Retain the 23 Nov interim interpretation with the final revision.
- Connect this readout to the proposal for QV.23.100.
- Preserve the revision suffix so the saved decision remains identifiable.

Final model-gate recommendation

- Redirect Corvenlith Therapeutics towards the reproducible receptor pathway.
- Qualify the inflammatory signature by its reduction after donor adjustment.
- Advance focused functional follow-up with donor-aware design.
- Dr T. Zawiernicki | 18 Dec 2023